Kickoff Session - Foundations of Deep Learning

1. Prerequisites and Motivation
2. Organizational Matters
3. Course Overview

Lecture Overview

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  - Even learning new algorithms
- Great convergence of AI around deep learning

Two Nobel Prizes in 2024 Went to AI Research

- Nobel Prize in Physics 2024:
  - John J. Hopfield and Geoffrey E. Hinton, “for foundational discoveries and inventions that enable machine learning with artificial neural networks”
- The Nobel Prize in Chemistry 2024:
  - David Baker, “for computational protein design”
  - Demis Hassabis and John M. Jumper, “for protein structure prediction”

Requirements and Relation to Other Courses

- Good knowledge of linear algebra and probability theory
- Good knowledge of machine learning, as, e.g., gained in our machine learning course
  - Possible to take in parallel, with some extra effort

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- Excerpt of machine learning courses for students starting this winter term
  - Term 1: Foundations: Machine learning, Deep learning, Reinforcement learning
  - Term 2: Specialized courses requiring these foundations: DL lab course, Advanced Deep Learning, Seminars, (AI), ...
  - Term 3: MSc project
  - Term 4: MSc thesis

Role of This Course in the Curriculum

- Several courses in Freiburg touch on deep learning a bit, but without focusing on it
  - Computer Vision
  - Machine Learning
  - Statistical Pattern Recognition
  - Information Retrieval
  - Artificial Intelligence
  - Introduction to Robotics

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  - Computer Vision
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  - Information Retrieval
  - Artificial Intelligence
  - Introduction to Robotics
- Deep learning is key for MSc projects / theses in many groups
  - Machine Learning (Hutter)
  - Robot Learning (Valada)
  - Computer Vision (Brox)
  - Neurorobotics (Boedecker)
  - Natural Language Processing (Bast)
  - AI for Oncology (Kalweit)
  - Generative Intelligence (Kortylewski)

Role of This Course in The Curriculum

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- For the practical component, we designed the Deep learning Lab course
  - Taught by several groups jointly in the summer term
  - Practical skills needed to do a project in deep learning
  - But not enough time to focus on foundations & deeper understanding

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  - Here: Focus on foundations & deep understanding
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  - In the lab course: practical skills
- This course is also a requirement for the Advanced Deep Learning course (summer term)

Foundations of Deep Learning and Advanced Deep Learning

- Week 1: Introduction to Deep Learning
- Week 2: Logistic Regression to MLPs
- Week 3: Backpropagation
- Week 4: Optimization
- Week 5: Regularization
- Week 6: Convolutional Neural Networks
- Week 7: Recurrent Neural Networks
- Week 8: Attention & Transformers
- Week 9: Practical Methodology
- Week 10: Auto-Encoders, VAEs, GANs
- Week 11: Uncertainty in Deep Learning
- Week 12: AutoML for Deep Learning

- Week 1: Introduction
- Week 2: Transformers I
- Week 3: Transformers II
- Week 4: Multidimensional Deep Learning
- Week 5: Graph Neural Networks
- Week 6: Similarity Learning
- Week 7: Multimodal Deep Learning
- Week 8: SSL and Foundation Models
- Week 9: Transfer, Adaptation, Continual
- Week 10: Generative Models
- Week 11: Neural Scene Representations and World Models
- Week 12: Industry Talks

Lecture Overview

1 Prerequisites and Motivation

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Organization: The Team



Prof. Abhinav Valada



Interim Prof. Steven Adriaensen



Mahmoud Safari



Imen Mahdi



Ayca Celik



Hagverdi Ibrahimli

Getting in touch: for almost everything, use ILIAS

If you do need to email, email to dl-orga-ws26@cs.uni-freiburg.de

Organization: Flipped Classroom Model

- Video lectures
- Weekly assignments
- In-person classroom discussions (Tuesdays, 14:15 - 15:45)

Organization: Flipped Classroom Model

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  - You watch these wherever/whenever you want
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  - Graded assignments, in groups of 3
  - 50% of assignment points required
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- In-person classroom discussions (Tuesdays, 14:15 - 15:45)
  - You need to watch the week's video lectures before
  - You should *ideally* start working on the week's assignment before
  - We discuss questions about the video lecture (also using slido)
  - We have a Kahoot quiz about the video lecture

Organization: Assignments

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  - Some implementations from scratch, to really understand the concepts
  - Rather than "just" use tool boxes

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- Let's form groups today!

Organization: Optional In-Person Slot for Working on Exercises

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- Attending these sessions is optional
  - No lecturing will take place!

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- An opportunity to
  - work on the exercises in class
  - get together as a team
  - interact with your tutors in person

Organization: ILIAS, Github, MOOC

- ILIAS
  - Video lectures
  - Slides / handouts
  - Links to external material (github, quizzes, slido, etc.)
  - Discussion forum
    - Please answer a question if you know the answer
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- MOOC: Massive Open Online Course
  - Free public version of this course is hosted on KI-Campus
  - Contains additional (interactive) quiz questions to self-test (OPTIONAL)

Organization: Amount of Work / ECTS points

- This course yields 6 ECTS points $\rightarrow$ 180 working hours
  - Video lectures: 12x1.5=18h
  - Flipped classroom slots: 14x2=28h
  - Self-study + assignments (optionally partly in-person): 12x7=84h
  - Exam preparation: 50h

- Time management options during the semester:

  A 11 hours/week (self-study, attend in-person sessions). RECOMMENDED

  B 7 hours/week (less self-study / interaction, more exam prep). FEASIBLE

  C Only shoot for 50% in the assignments. BAD IDEA

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- The “Pruefungsleistung” (PL) will be determined by a written exam
- The exam will...
  - take place on 17.03.2027 (time: TBA, duration: 90 minutes)
  - take the form of an ILIAS online test to be completed individually

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- Erasmus students who are not here during the exam
  - You can take the exam remotely at another university
  - You need to find out who is responsible for this at the other university and put us in touch by 26.01.2027
    - Please email to `dl-orga-ws26@cs.uni-freiburg.de` (do NOT email the individual profs)

Lecture Overview

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Course Overview

- Week 0: Kickoff!
- Week 1: Introduction to Deep Learning
- Week 2: From Logistic Regression to MLPs
- Week 3: Backpropagation
- Week 4: Optimization
- Week 5: Regularization
- Week 6: Convolutional Neural Networks (CNNs)
- Week 7: Recurrent Neural Networks (RNNs)
- Week 8: Attention & Transformers
- Christmas Break
- Week 9: Practical Methodology
- Week 10: Auto-Encoders, Variational Auto-Encoders, GANs
- Week 11: Uncertainty in Deep Learning
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Week 0: Kickoff

- Today's session!

Week 0: Kickoff

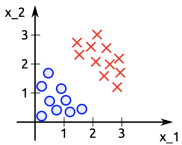
- Today's session!
- Catch up on Background Material (ILIAS)
  - Linear algebra
  - Bayesian linear regression
- Start working on the first exercise
  - Tests your background knowledge
  - Introduces Python and Git
  - Forming teams

Week 1: Introduction to Deep Learning

- Only few videos, to let you focus on setup for the exercises
- Motivation for deep learning
- Representation learning
- Flexibility of deep learning
- History of deep learning

Week 2: From Logistic Regression to MLPs

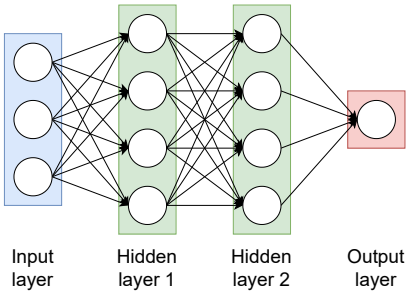
- The loss function we optimize: cross-entropy
- From standard ML to the simplest DL model
- Properties of multi-layer perceptrons



$$h_{\mathbf{w}}(\mathbf{x}) = g(\mathbf{w}_0 + \mathbf{w}_1\mathbf{x}_1 + \mathbf{w}_2\mathbf{x}_2)$$

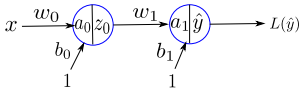
$$\mathbf{w} = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$$

$$J(h_{\mathbf{w}}(\mathbf{x}), y) = \begin{cases} -\log(h_{\mathbf{w}}(\mathbf{x})) & \text{for} \quad y = 1 \\ -\log(1 - h_{\mathbf{w}}(\mathbf{x})) & \text{for} \quad y = 0 \end{cases}$$



Week 3: Backpropagation

- Foundation for gradient-based optimization of loss function
- Basics of backpropagation
- Backpropagation in matrix form



$$\hat{y} = h_1(a_1) \qquad \frac{\partial L}{\partial \hat{y}}$$

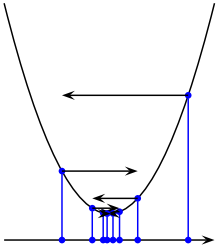
$$a_1 = z_0 w_1 + b_1 \qquad \frac{\partial L}{\partial a_1} = \frac{\partial L}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial a_1}$$

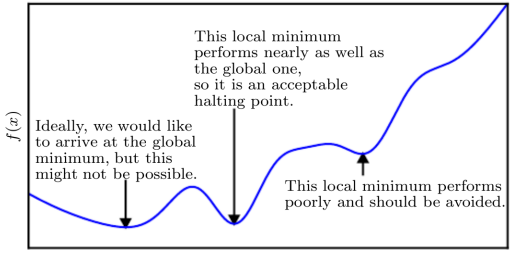
$$z_0 = h_0(a_0) \qquad \frac{\partial L}{\partial z_0} = \frac{\partial L}{\partial a_1} \frac{\partial a_1}{\partial z_0}$$

$$a_0 = x w_0 + b_0 \qquad \frac{\partial L}{\partial a_0} = \frac{\partial L}{\partial z_0} \frac{\partial z_0}{\partial a_0}$$

Week 4: Optimization for Deep Learning

- Non-convex optimization for DL
- Stochastic gradient descent (SGD)
- Learning rates, momentum
- Adaptive gradient algorithms, including Adam

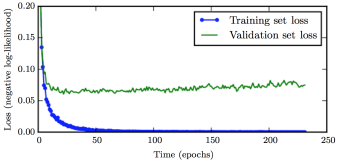


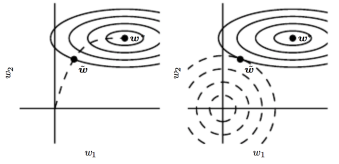


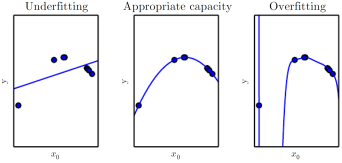
[Image Source: Goodfellow et al., 2016, p. 83, fig 4.3]

Week 5: Regularization

- Parameter sharing
- Dropout
- Data augmentation
- Preference for small weights



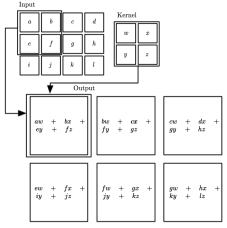


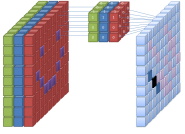


[Image Sources: Goodfellow et al., 2016, p. 243, fig 7.3, p. 247, fig 7.4, p.111, fig 5.2]

Week 6: Convolutional Neural Networks (CNNs)

- Convolutions
- Adding spatial context capability to neural nets

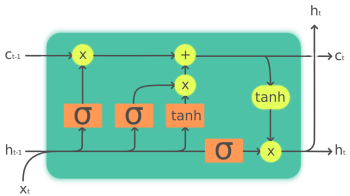




[Image Sources: Cecbur via Wikimedia Commons, Goodfellow et al., 2016, p330, fig 9.1]

Week 7: Recurrent Neural Networks (RNNs)

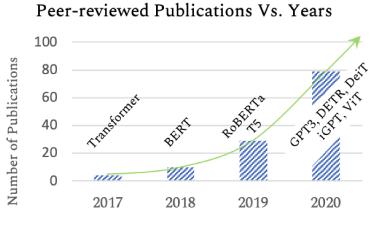
- Handling sequences
- Adding temporal context capability to neural nets
- LSTM

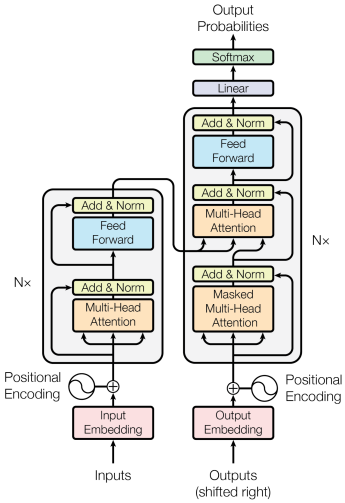


[Image Sources: Guillaume Chevalier via Wikimedia Commons, fdeloche via Wikimedia Commons]

Week 8: Attention & Transformers

- Learn to focus on certain parts of the input data, i.e. Attention Mechanism
- The Transformer architecture and recent variations, e.g. Vision-Transformers (ViT)
- Large, pretrained, Language Models, e.g. BERT





[Image Sources: Khan, Salman, et al. arXiv 2021, Vaswani, Ashish, et al. "Attention is all you need." NeurIPS 2017]

Week 9: Practical Methodology

- Practical methodology
- Avoiding issues / getting networks to train
- Special network architectures
- Self-supervised learning
- Manual HPO in Deep Learning

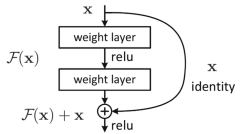
Input: Values of x over a mini-batch: $\mathcal{B} = \{x_{1...m}\}$;
Parameters to be learned: γ, β
Output: $\{y_i = \mathrm{BN}_{\gamma,\beta}(x_i)\}$

$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m} x_i \qquad \text{// mini-batch mean}$$

$$\sigma^2_{\mathcal{B}} \leftarrow \frac{1}{m}\sum_{i=1}^{m}(x_i - \mu_{\mathcal{B}})^2 \qquad \text{// mini-batch variance}$$

$$\widehat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma^2_{\mathcal{B}} + \epsilon}} \qquad \text{// normalize}$$

$$y_i \leftarrow \gamma \widehat{x}_i + \beta \equiv \mathrm{BN}_{\gamma,\beta}(x_i) \qquad \text{// scale and shift}$$

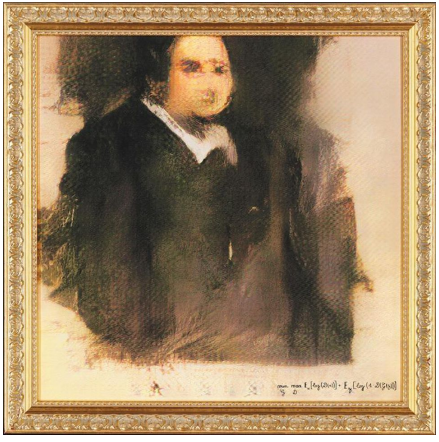


A residual block.

[Image Sources: Ioffe and Szegedy, 2015, He et al., 2016]

Week 10: Auto-Encoders, VAEs, GANs, Diffusion Models

- Variational autoencoders
- Generative adversarial networks
- Diffusion models

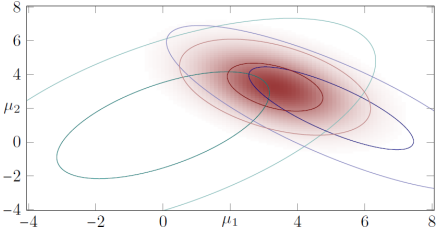


[Vincent, 2018]

Week 11: Uncertainty in Deep Learning

- Bayesian neural networks
- Approximate Bayesian Inference (variational inference, MCMC, prior-data fitted networks)
- Non-Bayesian estimates of uncertainty

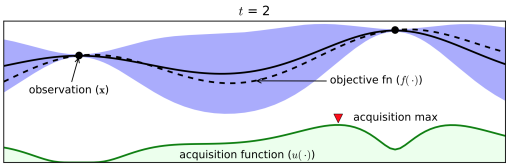




[Image Sources: Philipp Hennig; YavarPS via Wikimedia Commons]

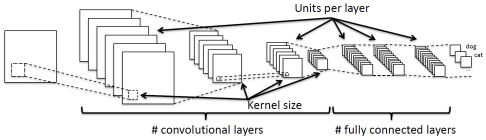
Week 12: AutoML for Deep Learning

- HPO in deep learning
- Blackbox optimization methods for HPO
- Speedup techniques for HPO



[Image Source: Brochu et al., 2010]

- Neural Architecture Search (NAS)



Question to Answer for Yourself / Discuss with Friends

- Your interests:
  Which part of the course are you most looking forward to?